UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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Lecture 26: RIP & Sparse Recovery

Content:

1. Restricted Isometry Property
2. Nullspace Property

Definition 0.1 (Restricted Isometry Property) The matrix $X \in \mathbb{R}^{n\times p}$ satisfies the RIP of order s and parameter $\delta \in (0,1)$, if for every s-sparse $\beta \in \mathbb{R}^p$

$$(1-\delta)\left\|\beta\right\|_2^2 \leq \left\|X\beta\right\|_2^2 \leq (1+\delta)\left\|\beta\right\|_2^2$$

the inequality above holds. We can also say that X is (s,δ)-RIP. Equivalently, X is (s,δ)-RIP iff $\|X_S^T X_S - I_S\|_{2\to 2} \leq \delta$.

Theorem 1.1 Suppose X satisfies RIP with order $2s$ and parameter $\delta = \delta_{2s} \in (0, 1/3)$. Then, for any s-sparse β^*, the unique solution to the following LP:

$$\min_{z\in\mathbb{R}^p} \left\|z\right\|_1, \qquad \text{subject to } Xz = X\beta^*$$

is just β^* itself.

Remark: LP is a convex program, which can be solved "easily" (e.g. by LASSO).

Lemma 1.2 For any s-sparse u, v with disjoint supports, we have

$$|\langle Xu, Xv\rangle| \le \delta_{2s} \|u\|_2 \|v\|_2$$

Proof: Given u, v have disjoint supports, we know $\langle u, v\rangle = 0$. Denote S as the union of the support of u and v, we have

$$\begin{aligned}
|\langle Xu, Xv\rangle| &= |\langle Xu, Xv\rangle - \langle u, v\rangle| = \left|\left\langle (X^TX - I)u, v\right\rangle\right| \\
&= \left|\left\langle (X_S^TX_S - I)u_S, v_S\right\rangle\right| \le \left\|(X_S^TX_S - I)u_S\right\|_2 \|v_S\|_2 \\
&\le \left\|X_S^TX_S - I\right\|_{2\to 2} \|u_S\|_2 \|v_S\|_2 \\
&\le \delta_{2s} \|u\|_2 \|v\|_2
\end{aligned}$$

Lemma 1.3 (Nullspace Property) Suppose for any $v \neq 0$ with $Xv = 0$ and any $S \subset [p]$ with $|S| = s$, we always have

$$\|v_S\|_1 < \frac{1}{2}\|v\|_1 \quad \Leftrightarrow \quad \|v_S\|_1 < \|v_{S^c}\|_1$$

Then for any s-sparse β^*, the unique solution to LP is β^*.

Proof: Fixed β that is s-sparse supported on $S \subset [p]$, $|S| = s$, take any $z \neq \beta$ with $Xz = X\beta = y$. Note that $z - \beta \neq 0$ and $X(z - \beta) = 0$. By assumption,

$$\|(z-\beta)_S\|_1 < \|(z-\beta)_{S^c}\|_1 \;\Rightarrow\; \|z_S - \beta\|_1 < \|z_{S^c}\|_1$$

then we have

$$\|\beta\|_1 \le \|\beta - z_S\|_1 + \|z_S\|_1 \le \|z_{S^c}\|_1 + \|z_S\|_1 = \|z\|_1$$

So for any z s.t. $Xz = X\beta$, $z \neq \beta$, we must have $\|z\|_1 > \|\beta\|_1$.

Proof of Theorem 1.1: Using Lemma 1.3, it suffices to show that for any z s.t. $Xz = 0$, $z \neq 0$, and any $S \subset [p]$, $|S| = s$, we can have $\|z_S\|_1 < \|z\|_1/2$. Consider a partition of $z \in \mathbb{R}^p$, $z = (z_1, \cdots, z_p)$, given their order

$$|z_{i_1}| \ge |z_{i_2}| \ge \cdots \ge |z_{i_p}|$$

Define the partition as follow:

$$S_0 = \{i_1, i_2, \cdots, i_S\}, \;\; S_1 = \{i_{S+1}, i_{S+2}, \cdots, i_{2S}\}, \;\; \cdots$$

Now it suffices to show that $\|z_{S^0}\| < \|z\|_1/2$.

As $z = z_{S_0} + z_{S_1} + \cdots$, $\{z_{S_i}\}_i$ have disjoint supports, and

$$Xz_{S_0} = -X(z_{S_1} + z_{S_2} + \cdots)$$

Moreover, $2s$ order RIP implies s order RIP, so by Lemma 1.2 we have

$$|\langle Xz_{S_0}, -Xz_{S_k}\rangle| \leq \delta_{2s} \|z_{S_0}\|_2 \|z_{S_k}\|_2$$

thus

$$\begin{aligned}
\|z_{S_0}\|_2^2 &\leq \frac{1}{1-\delta_{2s}} \|Xz_{S_0}\|_2^2 \\
&= \frac{1}{1-\delta_{2s}} \langle Xz_{S_0}, -X(z_{S_1} + z_{S_2} + \cdots)\rangle \\
&= \frac{1}{1-\delta_{2s}} \sum_{k\geq 1} \langle Xz_{S_0}, -Xz_{S_k}\rangle \\
&\leq \frac{\delta_{2s}}{1-\delta_{2s}} \|z_{S_0}\|_2 \sum_{k\geq 1} \|z_{S_k}\|_2
\end{aligned}$$

Dividing both sides by $\|z_{S_0}\|_2$, given $\delta_{2s} \in (0, 1, /3)$, we have:

$$\|z_{S_0}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \sum_{k\geq 1} \|z_{S_k}\|_2 \leq \frac{1}{2} \sum_{k\geq 1} \|z_{S_k}\|_2$$

To bound the L_2 norm on RHS with L_1 norm, we claim that $\|z_{S_k}\|_2 \leq \left\|z_{S_{k-1}}\right\|_1 / \sqrt{s}$. WLOG, assume $z_{S_{k-1}} = (a_1, \cdots, a_s)$, $z_{S_k} = (b_1, \cdots, b_s)$ with

$$|a_1| \geq \cdots \geq |a_s| \geq |b_1| \geq \cdots |b_2|$$

then

$$\|z_{S_k}\|_2 \leq \sqrt{s}|b_1| \leq \sqrt{s}|a_s| \leq \frac{|a_1| + \cdot + |a_s|}{\sqrt{s}} = \frac{\|z_{S_{k-1}}\|_1}{\sqrt{s}}$$

So given δ we have

$$\|z_{S_0}\|_2 \leq \frac{1}{2} \sum_{k\geq 1} \|z_{S_k}\|_2 \leq \frac{1}{2} \sum_{k\geq 1} \left\|z_{S_{k-1}}\right\|_1 \leq \frac{1}{2} \|z\|_1$$

So the setting of Theorem 1.1 satisfies the Nullspace property, which ensures that LP can exactly recover the original β^*.

Theorem 2.1 For any $\delta \in (0,1)$, there exists $c_1 = c_1(\delta),\ c_2 = c_2(\delta) > 0$ s.t. for every $1 \le s \le p$ and $\varepsilon > 0$, if

$$n \ge \frac{1}{c_2}\left(c_1 s \log\frac{pe}{s} + \log\frac{1}{\varepsilon}\right)$$

then the matrix $X/\sqrt{n}$ is (δ, s)-RIP with probability at least $1-\varepsilon$, where $X \in \mathbb{R}^{n\times p}$ has i.i.d. $\mathcal{N}(0,1)$ entries.

Remark:

- We can see $1/\sqrt{n}$ is the correct scaling because

$$\mathbb{E}\left\|\frac{1}{\sqrt{n}}X\beta\right\|_2^2 = \frac{1}{n}\beta^T\mathbb{E}[X^TX]\beta/n = \|\beta\|_2^2$$

- Alternatively, X' with i.i.d. $\mathcal{N}(0, 1/n)$ entries is (δ, s)-RIP.
- $n = \Omega(s\log(p/s))$ samples suffice for recovering the sparse regression setting.